

Department of CSE, UIU

CT-2 -- CSE 2217 – Fall22 -- Time 40mins -- Total Marks 20

Q1	Given an array of integers A = {-2, 1, -3, 7, -1, 2, 3, -5, 8}, find the <i>Maximum and Minimum</i> using divide-and-conquer . Show the necessary steps to support your answer.	5																					
Q2	You are given the following table containing symbols and their frequencies: <table><tr><td>Symbol</td><td>A</td><td>B</td><td>C</td><td>D</td><td>+</td></tr><tr><td>Frequency</td><td>40</td><td>10</td><td>20</td><td>15</td><td>15</td></tr></table> a) Generate the Huffman codes for the symbols. b) Decode 100010111001010 using the Huffman code that you generated.	Symbol	A	B	C	D	+	Frequency	40	10	20	15	15	5									
Symbol	A	B	C	D	+																		
Frequency	40	10	20	15	15																		
Q3	Solve the following recurrence relations using Master method: a) $T(n) = 2T\left(\frac{n}{2}\right) + \log n$ b) $T(n) = 27T(n/3)+n^3$	5																					
Q5	From the table below, consider the instance of the fractional knapsack problem with maximum capacity = 11. Specify which objects should be placed into the knapsack, and the maximum possible total profit. <table><tr><td></td><td>Object 1</td><td>Object 2</td><td>Object 3</td><td>Object 4</td><td>Object 5</td><td>Object 6</td></tr><tr><td>Weight</td><td>2</td><td>5</td><td>9</td><td>4</td><td>6</td><td>3</td></tr><tr><td>Profit</td><td>1</td><td>11</td><td>15</td><td>10</td><td>12</td><td>9</td></tr></table>		Object 1	Object 2	Object 3	Object 4	Object 5	Object 6	Weight	2	5	9	4	6	3	Profit	1	11	15	10	12	9	5
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